Simplify Trigonometric Expressions

Fundamental Identities			
Reciprocal Identities	$\csc x = \frac{1}{\sin x}$	$\sec x = \frac{1}{\cos x}$	$\cot x = \frac{1}{\tan x}$
	$\sin x = \frac{1}{\csc x}$	$\cos x = \frac{1}{\sec x}$	$\tan x = \frac{1}{\cot x}$
Quotient Identities	$\tan x = \frac{\sin x}{\cos x}$	$\cot x = \frac{\cos x}{\sin x}$	
Pythagorean Identities	$\sin^2 x + \cos^2 x = 1$	$\tan^2 x + 1 = \sec^2 x$	$1 + \cot^2 x = \csc^2 x$
Even and Odd Identities	$\sin(-x) = -\sin x$	$\tan(-x) = -\tan x$	$\cos(-x) = \cos x$
	$\csc(-x) = -\csc x$	$\cot(-x) = -\cot x$	$\sec(-x) = \sec x$

Simplify the expression. Write the final form with no fractions.

1. $\csc x \cos^2 x \sec x$

$= \frac{1}{\sin x} \cdot \cos x \cancel{\cos x} \cdot \frac{1}{\cancel{\cos x}}$

$= \frac{\cos x}{\sin x}$

$= \cot x$

2. $\frac{\cos^2 x}{\sin x \cot x}$

$= \frac{\cos x \cancel{\cos x}}{\cancel{\sin x} \cdot \frac{\cancel{\cos x}}{\cancel{\sin x}}}$

$= \cos x$

3. $\frac{\sec^2 x}{\tan^2 x} - 1$

$= \frac{\sec^2 x}{\tan^2 x} - \frac{\tan^2 x}{\tan^2 x}$

$= \frac{\sec^2 x - \tan^2 x}{\tan^2 x}$

$= \frac{1}{\tan^2 x}$

$= \cot^2 x$

4. $\frac{\cos x \csc^2 x - \cos x}{\cos x \csc x + \cos x}$

$= \frac{\cancel{\cos x}\ (\csc^2 x - 1)}{\cancel{\cos x}\ (\csc x + 1)}$

$= \frac{(\cancel{\csc x + 1})(\csc x - 1)}{\cancel{\csc x + 1}}$

$= \csc x - 1$

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Verify Trigonometric Identities

Guidelines for Verifying a Trigonometric Identity

1. Work with one side of the equation (usually the more complicated side) and keep the other side in mind as your final goal.

2. Look for opportunities to apply the fundamental identities.

   If the expression is a *product or quotient of factors*, consider the reciprocal and quotient identities.

   If *squared terms* are present, look to see if the terms can be grouped in one of the forms of a Pythagorean identity.

   If the expression involves a *negative argument*, consider using the even or odd function identities.

3. Apply basic algebraic techniques such as factoring, multiplying terms, combining like terms, and writing fractions with a common denominator.

4. Consider writing expressions explicitly in terms of sine and cosine.

Verify that the equation is an identity.

5. $\cos(-x)\sec(-x)+\tan(-x)\csc(-x)=\sec x+1$

$$\text{LHS: } \cos x\sec x+(-\tan x)(-\csc x)=\cancel{\cos x}\cdot\frac{1}{\cancel{\cos x}}+\frac{\cancel{\sin x}}{\cos x}\cdot\frac{1}{\cancel{\sin x}}$$

$$=1+\frac{1}{\cos x}=1+\sec x \quad \text{(RHS)}$$

6. $\dfrac{1}{\sec x-\tan x}-\dfrac{1}{\sec x+\tan x}=2\tan x$ $\quad \text{CD}=(\sec x-\tan x)(\sec x+\tan x)$

$$\text{LHS: } \frac{\sec x+\tan x-(\sec x-\tan x)}{(\sec x-\tan x)(\sec x+\tan x)}=\frac{\cancel{\sec x}+\tan x-\cancel{\sec x}+\tan x}{\sec^2 x-\tan^2 x}$$

$$=\frac{2\tan x}{1}=2\tan x \quad \text{(RHS)}$$

7. $\dfrac{\cos x}{\sec x+\tan x}=1-\sin x$

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$$\text{LHS: } \frac{\cos x}{\dfrac{1}{\cos x}+\dfrac{\sin x}{\cos x}}\cdot\frac{\cos x}{\cos x}=\frac{\cos^2 x}{1+\sin x}=\frac{1-\sin^2 x}{1+\sin x}=\frac{(1-\sin x)\cancel{(1+\sin x)}}{\cancel{1+\sin x}}=1-\sin x \quad \text{(RHS)}$$

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8. $\dfrac{\cot x}{\csc x - 1} = \dfrac{\csc x + 1}{\cot x}$

LHS: $\dfrac{\cot x}{\csc x - 1} \cdot \dfrac{\csc x + 1}{\csc x + 1} = \dfrac{\cot x(\csc x + 1)}{\csc^2 x - 1} = \dfrac{\cot x\,(\csc x + 1)}{\cot^2 x} = \dfrac{\csc x + 1}{\cot x}$ (RHS)

9. $\ln|\sin x| - \ln|\cos x| + \ln|\csc x| = \ln|\sec x|$

LHS: $\ln\left|\dfrac{\sin x \csc x}{\cos x}\right| = \ln\left|\dfrac{\sin x \cdot \dfrac{1}{\sin x}}{\cos x}\right| = \ln\left|\dfrac{1}{\cos x}\right| = \ln|\sec x|$ (RHS)

10. Verify that the equation is an identity by manipulating each side of the equation independently.

$$\frac{\sin x - \sin^3 x}{\cos^2 x} = \frac{\sin^2 x \sec x}{\tan x}$$

LHS

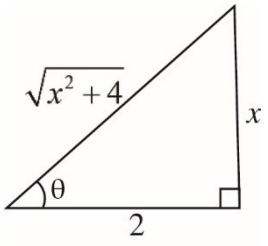
$$\frac{\sin x\left(1 - \sin^2 x\right)}{\cos^2 x}$$
$$= \frac{\sin x \cos^2 x}{\cos^2 x}$$
$$= \sin x$$

RHS

$$\frac{\sin^2 x \dfrac{1}{\cos x}}{\dfrac{\sin x}{\cos x}} \cdot \frac{\cos x}{\cos x}$$
$$= \frac{\sin^2 x}{\sin x}$$
$$= \sin x$$

Write an Algebraic Expression as a Trigonometric Expression

Some algebraic expressions may be written as trigonometric expressions. Note a few of the relationships in the right triangle.



$\tan\theta = \dfrac{x}{2} \rightarrow x = 2\tan\theta$ ($\dfrac{\text{opp}}{\text{adj}}$)

$\sec\theta = \dfrac{\sqrt{x^2+4}}{2} \rightarrow \sqrt{x^2+4} = 2\sec\theta$ ($\dfrac{\text{hyp}}{\text{adj}}$)

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$$0 < \theta < \frac{\pi}{2},$$

Write the given algebraic expression as a function of θ, where by making the given substitution.

11. $\sqrt{16 - x^2}$; substitute $x = 4\sin\theta$

$$\sqrt{16 - 16\sin^2\theta} = \sqrt{16\left(1 - \sin^2\theta\right)} = \sqrt{16\cos^2\theta} = 4\,|\cos\theta| = 4\cos\theta$$

↑ QI

12. $\dfrac{\sqrt{x^2 + 9}}{x}$; substitute $x = 3\tan\theta$

QI

$$\frac{\sqrt{9\tan^2\theta + 9}}{3\tan\theta} = \frac{\sqrt{9\left(\tan^2\theta + 1\right)}}{3\tan\theta} = \frac{\sqrt{9\sec^2\theta}}{3\tan\theta} = \frac{3\sec\theta}{3\tan\theta}$$

$$= \frac{\frac{1}{\cos\theta}}{\frac{\sin\theta}{\cos\theta}} = \frac{1}{\cancel{\cos\theta}} \cdot \frac{\cancel{\cos\theta}}{\sin\theta} = \frac{1}{\sin\theta} = \csc\theta$$

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